

The following are three additional equivalent definitions:

- For all polynomials, p , there exists a natural number, N , such that for all $\lambda > N$,

$$|f(\lambda)| \leq \frac{1}{|p(\lambda)|}. \quad \text{-----} \textcircled{1}$$

- For all polynomials, p , there exists a natural number, N , such that for all $\lambda > N$,

$$\frac{1}{|f(\lambda)|} \geq |p(\lambda)|. \quad \text{-----} \textcircled{2}$$

- For all natural numbers, c , there exists a natural number, N , such that for all $\lambda > N$,

$$\frac{1}{|f(\lambda)|} \geq |\lambda^c|. \quad \text{-----} \textcircled{3}$$

Exercise 1. Prove that a function, f satisfying any one of the above properties satisfies all of the above properties.

We prove,

$$\textcircled{1} \iff \textcircled{2} \iff \textcircled{3}$$

i) $\textcircled{1} \iff \textcircled{2}$

Let f satisfy $\textcircled{1}$.

i.e., $\forall$ polynomial $p(\lambda)$, $\exists N \in \mathbb{N}$ such that

$$\forall \lambda > N, |f(\lambda)| \leq \frac{1}{|p(\lambda)|}$$

Since $f(\lambda) \neq 0$ for all sufficiently large λ ,

$$\textcircled{1} \Rightarrow \frac{1}{|f(\lambda)|} \geq |p(\lambda)|$$

Thus $\textcircled{2}$ holds with same witness N .

$$\text{Hence } \textcircled{1} \Rightarrow \textcircled{2} \text{ -----} \textcircled{4}$$

Conversely if f satisfy $\textcircled{2}$,

Multiplying both the sides of the inequality by $|f(\lambda)|$, we get

$$|f(\lambda)| \leq \frac{1}{|p(\lambda)|}$$

$$\text{So } \textcircled{1} \text{ holds. Hence } \textcircled{2} \Rightarrow \textcircled{1} \text{ -----} \textcircled{5}$$

From $\textcircled{4}$ & $\textcircled{5}$, we conclude

$$\textcircled{1} \iff \textcircled{2}$$

ii) ② $\Leftrightarrow$ ③

Fix any $c \in \mathbb{N}$

Define a polynomial $p(\lambda) = \lambda^c$

Since f satisfies ②, $\exists N \in \mathbb{N}$ such that

$$\forall \lambda > N, \frac{1}{|f(\lambda)|} \geq |p(\lambda)| = |\lambda^c|$$

Hence ② $\Rightarrow$ ③ ————— ⑥

Conversely,

let $p(\lambda) = a_d \lambda^d + a_{d-1} \lambda^{d-1} + \dots + a_0$ with degree d and leading coefficients $a_d > 0$.

To find $N \in \mathbb{N}$ such that $\forall \lambda > N, \frac{1}{|f(\lambda)|} \geq |p(\lambda)|$

By standard bounds on polynomial growth, $\exists C > 0$ and $\lambda_0 \in \mathbb{N}$ such that $\forall \lambda > \lambda_0, |p(\lambda)| \leq C \lambda^d$ ————— ⑦

Let $c = d+1 \in \mathbb{N}$.

then from ③, $\exists N_1 \in \mathbb{N}$ such that

$$\forall \lambda > N_1, \frac{1}{|f(\lambda)|} \geq \lambda^c = \lambda^{d+1} \text{ ————— ⑧}$$

From ⑥ and ⑦, $\forall \lambda > \max\{N_1, \lambda_0\}$, we have,

$$\frac{1}{|f(\lambda)|} \geq \lambda^{d+1} \geq C \lambda^d \geq |p(\lambda)|$$

$$\Rightarrow \frac{1}{|f(\lambda)|} \geq |p(\lambda)|$$

So f satisfies ②

Hence ③ $\Rightarrow$ ② ————— ⑨

$\therefore$ From ⑥ and ⑨ together, we have

$$\text{②} \Leftrightarrow \text{③}$$

Since we have $① \iff ②$ and $② \iff ③$,
it follows by transitivity that

$$① \iff ② \iff ③$$

Thus any function f satisfying one of the conditions
satisfies all three //

Which of the following are negligible functions in λ ? Justify your answers.

$$\frac{1}{2^{\lambda/2}} \quad \frac{1}{2^{\log(\lambda^2)}} \quad \frac{1}{\lambda^{\log(\lambda)}} \quad \frac{1}{\lambda^2} \quad \frac{1}{2^{(\log \lambda)^2}} \quad \frac{1}{(\log \lambda)^2} \quad \frac{1}{\lambda^{1/\lambda}} \quad \frac{1}{\sqrt{\lambda}} \quad \frac{1}{2^{\sqrt{\lambda}}}$$

A function $f(\lambda)$ is negligible if, $\forall$ constant $c > 0$,

$$\lim_{\lambda \rightarrow \infty} \lambda^c \cdot f(\lambda) = 0$$

1) $\frac{1}{2^{\lambda/2}}$

We need,

$$\lim_{\lambda \rightarrow \infty} \lambda^c \cdot \frac{1}{2^{\lambda/2}} = 0 \quad \forall c.$$

Using the identity, $\lambda^c = 2^{c \log_2 \lambda}$ [$\because 2^{\log_2 \lambda} = \lambda$], we have

$$\frac{\lambda^c}{2^{\lambda/2}} = 2^{c \log_2 \lambda - \lambda/2}$$

As $\lambda \rightarrow \infty$, $c \log_2 \lambda - \lambda/2 \rightarrow -\infty$ since $\lambda/2$ term dominates

$$\Rightarrow 2^{c \log_2 \lambda - \lambda/2} \rightarrow 0$$

$\therefore \frac{1}{2^{\lambda/2}}$ is negligible.

2) $\frac{1}{2^{\log \lambda^2}}$

$$\Rightarrow \frac{1}{2^{\log \lambda^2}} = \frac{1}{2^{2 \log \lambda}} = \frac{1}{(2^{\log \lambda})^2} = \frac{1}{\lambda^2}$$

Considering $c=3$, then

$$\lim_{\lambda \rightarrow \infty} \lambda^3 \cdot \frac{1}{\lambda^2} = \lambda \rightarrow \infty \quad \therefore \text{Not negligible.}$$

$$3) \frac{1}{\lambda^{\log \lambda}}$$

$$\lambda^c \cdot \frac{1}{\lambda^{\log \lambda}} = \lambda^{c - \log \lambda}$$

As $\lambda \rightarrow \infty$, $\log \lambda \rightarrow \infty$ so $c - \log \lambda \rightarrow -\infty$

$$\therefore \lim_{\lambda \rightarrow \infty} \lambda^c \cdot \frac{1}{\lambda^{\log \lambda}} \rightarrow 0 \quad \therefore \text{Negligible.}$$

$$4) \frac{1}{\lambda^2}$$

Considering $c=3$, then

$$\lim_{\lambda \rightarrow \infty} \lambda^3 \cdot \frac{1}{\lambda^2} = \lambda \rightarrow \infty \quad \therefore \text{Not negligible.}$$

$$5) \frac{1}{2^{(\log \lambda)^2}}$$

$$\text{Using } \lambda^c = 2^{c \log_2 \lambda} \Rightarrow \lambda^c \cdot \frac{1}{2^{(\log \lambda)^2}} = \frac{2^{c \log_2 \lambda}}{2^{(\log_2 \lambda)^2}}$$

$$\text{As } \lambda \rightarrow \infty, \quad c \log_2 \lambda - \log_2 \lambda^2 \rightarrow -\infty \text{ as } (\log_2 \lambda)^2 \text{ dominates.}$$

$$\therefore 2^{c \log_2 \lambda - \log_2 \lambda^2} \rightarrow 0 \quad \therefore \text{Negligible.}$$

$$6) \frac{1}{(\log_2 \lambda)^2}$$

Considering $c=2$, we have

$$\lim_{n \rightarrow \infty} \lambda^c f(\lambda) = \lim_{n \rightarrow \infty} \lambda^2 / (\log_2 \lambda)^2 = \lim_{n \rightarrow \infty} \left(\frac{\lambda}{\log_2 \lambda} \right)^2 = \infty$$

as λ^2 term dominates $(\log_2 \lambda)^2$ term

$\therefore$ Not negligible.

$$7) \frac{1}{\lambda^{1/2}}$$

$$\text{Let } f(\lambda) = \lambda^{1/2} \Rightarrow \ln f(\lambda) = \frac{1}{2} \ln \lambda$$

$$\Rightarrow \lim_{\lambda \rightarrow \infty} \frac{\ln \lambda}{\lambda} = 0$$

$$\therefore \lim_{\lambda \rightarrow \infty} \ln f(\lambda) = 0$$

Raising exponentially, we get

$$\lim_{\lambda \rightarrow \infty} e^{\ln f(\lambda)} = e^0 = 1$$

$$\text{So } \lim_{\lambda \rightarrow \infty} \frac{1}{\lambda^{1/2}} = 1$$

$$\therefore \lim_{\lambda \rightarrow \infty} \lambda^c f(\lambda) = \lim_{\lambda \rightarrow \infty} \lambda^c \cdot \frac{1}{\lambda^{1/2}} = \infty$$

Because as $\lambda \rightarrow \infty$, $\frac{1}{\lambda^{1/2}} \rightarrow 1$ so $\lambda^c \rightarrow \infty$

$\therefore$ Not negligible.

$$8) \frac{1}{\sqrt{\lambda}} = \frac{1}{\lambda^{1/2}}$$

considering $c=1$, we have

$$\lim_{\lambda \rightarrow \infty} \lambda^c \cdot f(\lambda) = \lim_{\lambda \rightarrow \infty} \lambda \frac{1}{\lambda^{1/2}} = \infty \text{ as } \lambda \text{ term dominates } \frac{1}{\sqrt{\lambda}}$$

$\therefore$ Not negligible

$$9) \frac{1}{2^{\sqrt{\lambda}}}$$

$$\begin{aligned} \lim_{\lambda \rightarrow \infty} \lambda^c f(\lambda) &= \lim_{\lambda \rightarrow \infty} 2^{c \log_2 \lambda} \cdot \frac{1}{2^{\sqrt{\lambda}}} \\ &= \lim_{\lambda \rightarrow \infty} 2^{c \log_2 \lambda - \sqrt{\lambda}} \end{aligned}$$

since $\sqrt{\lambda} \gg \log_2 \lambda$,

$$c \log_2 \lambda - \sqrt{\lambda} \rightarrow -\infty \text{ as } \lambda \rightarrow \infty$$

$$\therefore \lim_{\lambda \rightarrow \infty} 2^{c \log_2 \lambda - \sqrt{\lambda}} = 2^{-\infty} = 0$$

$\therefore$ Negligible.

Suppose f and g are negligible.

(a) Show that $f + g$ is negligible.

(b) Show that $f \cdot g$ is negligible.

a) Given f and g are negligible.

To show $f + g$ is negligible.

Let $c \in \mathbb{N}$, then,

$$\lim_{\lambda \rightarrow \infty} \lambda^c (f(\lambda) + g(\lambda)) = \lim_{\lambda \rightarrow \infty} \lambda^c f(\lambda) + \lim_{\lambda \rightarrow \infty} \lambda^c g(\lambda)$$

Since f and g are negligible, we have

$$\lim_{\lambda \rightarrow \infty} \lambda^c f(\lambda) = \lim_{\lambda \rightarrow \infty} \lambda^c g(\lambda) = 0$$

$$\therefore \lim_{\lambda \rightarrow \infty} \lambda^c (f(\lambda) + g(\lambda)) = 0 + 0 = 0$$

Hence $f + g$ is negligible.

b) Let $c \in \mathbb{N}$, since f and g are negligible, then for any $c \nexists c_1, c_2 \in \mathbb{N}$ such that

$$\lambda^{c_1} f(\lambda) \rightarrow 0 \text{ and } \lambda^{c_2} g(\lambda) \rightarrow 0 \text{ as } \lambda \rightarrow \infty$$

Let $c = c_1 + c_2$, then

$$\begin{aligned} \lambda^c f(\lambda) g(\lambda) &= \lambda^{c_1 + c_2} f(\lambda) g(\lambda) \\ &= \lambda^{c_1} f(\lambda) \cdot \lambda^{c_2} g(\lambda) \rightarrow 0 \cdot 0 = 0 \\ &\quad \text{as } \lambda \rightarrow \infty \end{aligned}$$

$\therefore f \cdot g$ is negligible.

Show that when f is negligible, then for every polynomial p , the function $p(\lambda)f(\lambda)$ not only approaches 0, but it is also negligible itself.

Suppose there exist a polynomial $p(\lambda)$ such that $p(\lambda) \cdot f(\lambda)$ is not negligible.

i.e., $\exists$ some polynomial $q(\lambda)$ such that
$$\lim_{\lambda \rightarrow \infty} q(\lambda) \cdot |p(\lambda)f(\lambda)| \neq 0 \quad \text{--- ①}$$

Let $u(\lambda) = q(\lambda)p(\lambda)$ [$\because p$ and q are polynomials, their product u is also a polynomial]

Consider,

$$u(\lambda) \cdot |f(\lambda)| = q(\lambda)p(\lambda) \cdot |f(\lambda)|$$

Based on our assumption ①, we have

$$\lim_{\lambda \rightarrow \infty} u(\lambda) \cdot |f(\lambda)| \neq 0$$

Thus f is not negligible.

This proves the contrapositive.

Hence the original statement is true.

$\therefore$ If f is negligible, then $\forall p, p(x)f(x)$ is also negligible.

Prove that the $\approx$ relation is transitive. Let $f, g, h : \mathbb{N} \rightarrow \mathbb{R}$ be functions. Using the definition of the $\approx$ relation, prove that if $f \approx g$ and $g \approx h$ then $f \approx h$. You may find it useful to invoke the *triangle inequality*: $|a - c| \leq |a - b| + |b - c|$.

To prove, if $f \approx g$ and $g \approx h$, then $f \approx h$.

By definition, $f \approx g \iff |f(\lambda) - g(\lambda)|$ is negligible.

We also have,

$$f \approx g \Rightarrow |f - g| \text{ is negligible.}$$

$$g \approx h \Rightarrow |g - h| \text{ is negligible.}$$

We want to show

$$f \approx h \Rightarrow |f - h| \text{ is negligible.}$$

Using triangle inequality,

$$|f(\lambda) - h(\lambda)| \leq |f(\lambda) - g(\lambda)| + |g(\lambda) - h(\lambda)|$$

Since $|f - g|$ and $|g - h|$ is negligible.

Then their sum is also negligible

$$\text{Hence } |f(\lambda) - h(\lambda)| \leq \text{negligible} + \text{negligible}$$

$$\text{so } |f - h| \text{ is negligible.}$$

$\therefore$ If $f \approx g$ and $g \approx h$, then $f \approx h$

Thus $\approx$ is transitive. //

Exercise 4. Let (G, S, V) be a secure signature scheme with the message space $\{0, 1\}^\lambda$. Let $\hat{G}$ be a new key generation function that simply calls G twice and returns a pair of secret keys, (sk_0, sk_1) , and the corresponding pair of public keys, (pk_0, pk_1) . Which of the following signature schemes, $(\hat{G}, \hat{S}, \hat{V})$, are secure? Show an attack or prove security (by reduction).

(a) Prove that "Sign halves" is secure:

$$\hat{S}((sk_0, sk_1), (m_L, m_R)) = (S(sk_0, m_L), S(sk_1, m_R)),$$

$$\hat{V}((pk_0, pk_1), (m_L, m_R), (\sigma_0, \sigma_1)) = V(pk_0, m_L, \sigma_0) \wedge V(pk_1, m_R, \sigma_1).$$

Let $\Pi = (G, S, V)$ be a digital signature scheme over message space $\mathcal{M} = \{0, 1\}^\lambda$. It is EUF-CMA secure if for all PPT adversaries A , the probability that A outputs a valid signature on a new message is negligible.

$$\text{i.e., } p_H \left[\begin{array}{l} (pk, sk) \leftarrow G(1^\lambda); \\ (m^*, \sigma^*) \leftarrow A^{S(sk, \cdot)}(pk); \\ V(pk, m^*, \sigma^*) = 1 \wedge m^* \notin Q \end{array} \right] \leq \text{negl}(\lambda)$$

where Q is the set of messages queried to the signing oracle.

a) Given,

Let the message space be $\mathcal{M}' = \mathcal{M} \times \mathcal{M}$

Signing be $\hat{S}((sk_0, sk_1), (m_L, m_R)) = (S(sk_0, m_L), S(sk_1, m_R))$

& verification $\hat{V}((pk_0, pk_1), (m_L, m_R), (\sigma_0, \sigma_1)) = V(pk_0, m_L, \sigma_0) \wedge V(pk_1, m_R, \sigma_1)$

Let adversary A do

- 1) Query signature on
 $(m_L, m_R) \rightarrow (\sigma_L, \sigma_R)$ and $(m_L', m_R) \rightarrow (\sigma_L', \sigma_R)$
- 2) Produce forgery on new message.

$$(m_L, m_R), (\sigma_L, \sigma_R)$$

Now, $m_L \in \mathcal{Q}_L, m_R \in \mathcal{Q}_R$ but $(m_L, m_R) \notin \mathcal{Q}$
So this is a new message in $\mathcal{M}'$, yet

$$\hat{V} = (CPK_0, PK_1, (m_L, m_R), (\sigma_L, \sigma_R)) = 1$$

because

$$V(PK_0, m_L, \sigma_L) = 1, V(PK_1, m_R, \sigma_R) = 1$$

$\therefore$ A forges a valid signature on a new message (m_L, m_R)
by reusing old parts.

$\therefore$ Sign halves is not secure (EUF-CMA is broken)

(b) Show that "Sign with randomness" is **not** secure:

$$\hat{S}(sk_0, m) = [\text{choose random } r \in \{0, 1\}^\lambda; \text{output } (r, S(sk_0, m \oplus r), S(sk_0, r))],$$

$$\hat{V}(pk_0, m, (r, \sigma_0, \sigma_1)) = V(pk_0, m \oplus r, \sigma_0) \wedge V(pk_0, r, \sigma_1).$$

Let signing be defined as

$$\hat{S}(sk_0, m) = (\kappa, \sigma_0, \sigma_1) \text{ where } \kappa \leftarrow \{0, 1\}^\lambda,$$

$$\sigma_0 = S(sk_0, m \oplus \kappa), \sigma_1 = S(sk_0, \kappa)$$

{ verification

$$\hat{V}(pk_0, m, (\kappa, \sigma_0, \sigma_1)) = V(pk_0, m \oplus \kappa, \sigma_0) \wedge V(pk_0, \kappa, \sigma_1)$$

Let A query twice:

$$1. m_1 \rightarrow (\kappa_1, \sigma_{1,0}, \sigma_{1,1})$$

$$2. m_2 \rightarrow (\kappa_2, \sigma_{2,0}, \sigma_{2,1})$$

and construct forgery on: $m^* = \kappa_1 \oplus (m_2 \oplus \kappa_2),$

Then,

$$(\tilde{\kappa} = \kappa_1, \tilde{\sigma}_0 = \sigma_{2,0}, \tilde{\sigma}_1 = \sigma_{1,1})$$

$$m^* \oplus \tilde{\kappa} = \kappa_1 \oplus (m_2 \oplus \kappa_2) \oplus \kappa_1,$$

$$= m_2 \oplus \kappa_2. \text{ so } \tilde{\sigma}_0 \text{ is valid.}$$

{ $\tilde{\kappa} = \kappa_1$ so $\tilde{\sigma}_1$ is valid

$$\text{So, } \hat{V}(pk_0, m^*, (\kappa_1, \sigma_{2,0}, \sigma_{1,1})) = 1$$

$\Rightarrow$ Adversary forges signature on a new message
 $m^* \notin \mathcal{Q}$

$\therefore$ Sign with randomness is not secure.

(c) Prove that "Accept one valid" is secure:

$$\hat{S}((sk_0, sk_1), m) = (S(sk_0, m), S(sk_1, m)),$$

$$\hat{V}((pk_0, pk_1), m, (\sigma_0, \sigma_1)) = V(pk_0, m, \sigma_0) \vee V(pk_1, m, \sigma_1).$$

(Hint: Consider using that $\Pr[\mathcal{A} \text{ wins}] = \Pr[\sigma_0 \text{ valid}] + \Pr[\sigma_1 \text{ valid}] - \Pr[\sigma_0, \sigma_1 \text{ both valid}]$, and that the last part must be negligible).

Let the signing be

$$\hat{S}((sk_0, sk_1), m) = (S(sk_0, m), S(sk_1, m)) = (\sigma_0, \sigma_1)$$

& verification be

$$\hat{V}((pk_0, pk_1), m, (\sigma_0, \sigma_1)) = V(pk_0, m, \sigma_0) \vee V(pk_1, m, \sigma_1)$$

Assume $\exists$ adversary A breaking EUF-CMA of $\hat{S}$ with non negligible probability ϵ .

i.e., A signs a new m^* successfully with probability $\epsilon(\lambda)$

$$\Pr \left[\hat{V}((pk_0, pk_1), m^*, (\sigma_0^*, \sigma_1^*)) = 1 \right] \geq \epsilon$$

We build B , breaking the original scheme (G, S, V) :

- Input: B receives pk from challenger and randomly picks $b \in \{0, 1\}$
- Set $(pk_b, pk_{1-b}) = (pk, \text{fresh})$, B gives (pk_0, pk_1) to A
- On any signing query m , B queries its signing oracle to produce σ_b and locally compute σ_{1-b} , then returns (σ_0, σ_1) to A .
- When A outputs $(m^*, (\sigma_0^*, \sigma_1^*))$:

If $V(pk, m^*, \sigma_b^*) = 1$ and m^* was never queried,

B outputs (m^*, σ_b^*) as the forgery.

We have

$$\begin{aligned} P_A[A \text{ wins}] &= P_A[\sigma_0^* \text{ valid}] + P_A[\sigma_1^* \text{ valid}] \\ &\quad - P_A[\sigma_0^* \sigma_1^* \text{ both valid}] \end{aligned}$$

Let $\epsilon_0 = P_A[\sigma_0^* \text{ valid}]$, $\epsilon_1 = P_A[\sigma_1^* \text{ valid}]$

Then

$$\epsilon \leq \epsilon_0 + \epsilon_1$$

So at least one of ϵ_0 or ϵ_1 is $\geq \epsilon/2$ which is non-negligible. Thus B succeeds with non negligible probability.

This contradicts our assumed security.

$\therefore$ Accept one valid is EUF-CMA secure.